

2023

(Session : 2022-25)

(Paper ID : 11202)

Time : 3 hours

Full Marks : 80

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. Draw the Block Diagram of CPU. Explain the structure and functioning of Arithmetic Logic Unit (ALU).
2. What is Demorgan's Theorem? Explain it with diagram and truth table.
3. What do you mean by addressing mode? Discuss any four Addressing Modes.
4. What is Memory? Explain different types of memory.
5. What do you mean by Digital counter? Design a 3-bit binary counter using Flip-Flop.
6. What is Binary Arithmetic? Explain Binary Addition and Subtraction with example.
7. What is Multiplexer and De-Multiplexer? Explain it.
8. Explain is Logic Gates? Explain it with diagram and truth table.
9. What is complement? To find the 1's and 2's complements of following:
 - (a) $(10101011)_2$
 - (b) $(11100011)_2$
 - (c) $(11010010)_2$
 - (d) $(10101010)_2$
10. Write short notes on any two of the following:
 - (a) Sequential Circuit
 - (b) Universal gates
 - (c) ROM
 - (d) Register